

강의계획서

2025학년도 2학기

과목명	이산구조		교과목 번호	MC3202	학점	3	강의실	창708
			교과 구분	AP 과목	시수	3		
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1. 교과목 개요

1.1 Course Objectives

On completion of this course, you'll hopefully be able to:

- Think in the language of mathematics (i.e. set/relation/logic)
 - Throughout this course, you will be trained to be able to think in terms of set/relation/logic fairly freely and comfortably.
- Synthesize (heavy) mathematical proofs;
- Model/analyze real-world problems and design algorithmic solutions to the problems along with mathematical proofs using discrete structures and combinatorial methods
- Think computationally/algorithmically;
 - appreciate the importance of algorithms in CS and beyond;
 - understand/appreciate limits of computation.

1.2 Contents

- Logic, Proofs, Sets, Relations, Algebraic structures (groups, rings, fields, lattices, etc.)
- Graph: Basic notions, Isomorphism, Coloring, Planarity, Connectivity, Independence
- Theory of computation: Turing machines, Undecidability, Incompleteness, Intractability
- Modular arithmetic, Cryptography
- Combinatorial analysis
 - Lattice paths, Cayley's formula, Pölya/Burnside methods, Chains & Antichains
- Combinatorial designs
 - Finite Fields, Finite geometries, Latin squares, block designs
- Combinatorial optimizations
 - Graph algorithms: shortest paths, minimum spanning trees, maximum flows
 - Matroids, Algebraic path problems
- Game theory
 - Combinatorial games: Nim, Sprague-Grundy functions, surreal numbers
 - Classical games: Nash equilibrium, solution concepts, mechanism design

2. 중점 목표 역량

교과 핵심 역량			
problem-solving skills	logical thinking	problem discovery and definition	creativity

3. 교재 및 참고도서

가. 주교재 : 자체 제작 수업 자료

나. 참고 서적

- “Discrete and Combinatorial Mathematics”, R. Grimaldi
- “Extremal Combinatorics”, S. Jukna
- “Handbook of Discrete and Combinatorial Mathematics”, K. Rosen et al.
- “Mathematical Logic”, H. Schwichtenberg
- “Introduction to Set Theory”, K. Hrbacek and T. Jech
- “On Numbers and Games”, J. Conway
- “A Course in Game Theory”, M. Osborne and A. Rubinstein
- “Graph Theory”, R. Diestel
- “The Probabilistic Method”, N. Alon and J. Spencer
- “Introduction to Theory of Computation”, M. Sipser
- “Introduction to Modern Cryptography”, J. Katz and Y. Lindell
- “Computers and Intractability: A Guide to the Theory of NP-Completeness”, M. Garey et al.
- “Algorithm Design”, J. Kleinberg and E. Tardos
- “Introduction to Algorithms”, T. Cormen, C. Leiserson, and R. Rivest

4. 수업 방법 :

가. 강의

5. 평가 기준 및 방법

가. 평가 기준

과제 (14회)	퀴즈 (수시)	기말고사	출결 및 태도	합계
10%	10%	70%	10%	100%

나. 평가 방법

- 14 problem sets
 - Scoring policy: full score if 2/3 submitted, otherwise 0 point (refer to lec00_overview.pdf)
- Final exam: 9:00-23:00 on the Saturday right before the final exam period (2025/12/6)
- Late-work policy: not accepted
- Make-up final exam for officially approved absence (e.g. COVID, illness, out-of-school competition, conference)
 - immediately after the day all students are released from quarantine/...
 - 0 point if do not show up on the make-up final test

6. 학점 부여 기준

- 출결 및 태도: 총 수업 시간의 3/4 이상 부여시 100% 부여
- 병결/공결 등으로 정규 기말고사에 미응시 경우: 대체시험 실시하여 그대로 기말고사 점수로 반영
- 학점 부여 기준: KAIST/POSTECH AP 담당 교수의 피드백으로 결정
- 과목우수상 기준: 원점수 기준으로 정렬하여 결정

7. 주별 강의 계획

Lec #	Topics	Assignments	Categories
1	Sets/Logic (basic notions)		Sets/Logic
2	Subgroups, Permutation Groups		Algebraic Structures
3	Group Actions/Homomorphisms	PS #1 due	
4	Groups & Rubik's Cube		
5	Finite Fields	PS #2 due	
6	Finite Geometries		Relations
7	Order/Equivalence Relations	PS #3 due	
8	Operations/Relations on Relations		Graphs
9	Graphs (basic notions)	PS #4 due	
10**	Graphs problems in CS		
11	Trees		Combinatorial Analysis
12	Counting	PS #5 due	
13	Finite Probability Space		
14	Cayley's Formula		
15-16	Chains/Antichains	PS #6 due	Algebraic Str.
17	Lattices (Stable Matching)		
18**	Turing Machines & Algorithms		Theory of Computation
19**	Universality & Undecidability	PS #7 due	
20	Incompleteness Theorems		
21	<i>Wrap-up for the midterm</i>	PS #8 due	
22*	Shortest Paths (Dijkstra)		Graph Algorithms (Combinatorial Optimization)
23	Shortest Paths (Floyd-Warshall)		
24	Algebraic Path Problems (Semiring)	PS #9 due	
25*	Minimum Spanning Trees		
26	Matroids		
27*	Flow Networks	PS #10 due	Intractability
28*	Polynomial-Time Reductions		
29*	NP-Completeness		Game Theory
30	Combinatorial Games	PS #11 due	
31	Nash Equilibrium		
32	Mechanism Design		Combinatorial Design
33	Latin Squares	PS #12 due	
34	Block Designs		Cryptography
35**	Divisibility		
36**	Modular Arithmetic	PS #13 due	
37**	Cryptology (informal overview)		
38**	Public-Key Cryptosystems		
39	<i>Wrap-up for the final exam</i>	PS #14 due	

주(기간)	주제 및 내용	(비고)
제1주 8.18.~8.22.	1. Sets/Logic (Set/Logic) - Propositional/predicate logic - Logical systems, soundness/completeness/consistency - Cardinalities of infinite sets, Russell's paradox - ZFC axioms 2. Groups & Rubik's cube #1 (Algebraic structures) - Subgroups, generators - The Symmetric group, disjoint cycle decomposition - Group representation for Rubik's cube 3. Groups & Rubik's cube #2 (Algebraic structures) - Group homomorphisms, kernels - The sign homomorphism, The alternating group, - Group actions, orbits	
핵심역량	problem discovery and definition	
제2주 8.25.~8.29.	4. Groups & Rubik's cube #3 (Algebraic structures) - Necessary condition for valid configurations - Sufficient condition (by using conjugators and generators) - Number of valid configurations - A lower bound for the worst-case number of moves 5. Finite fields (Algebraic structures) - Prime fields, polynomial rings - Irreducible polynomials, division of polynomials - The Galois field, uniqueness (up to isomorphism) 6. Finite geometries (Combinatorial design) - Axioms of Euclidean geometry - Affine planes, construction from the finite field - Projective planes	
핵심역량	problem-solving skills, logical thinking, creativity	
제3주 9.1.~9.5.	7. Relations #1 (Relations) - Operations on relations - Properties of binary relations - Order relations 8. Relations #2 (Relations) - Equivalence relations and partitions - Partition refinement, relations on relations 9. Graphs #1 (Graphs) - Basic notions, adjacency - Connectivity, articulation points - Eulerian/Hamiltonian paths/cycles - Planarity testing, Kuratowski's theorem - Graph isomorphism, subgraph isomorphism	
핵심역량	problem-solving skills, logical thinking, creativity	
제4주 9.8.~9.12.	10. Graphs #2 (Graphs) - Shortest paths, MSTs - Max-flow/min-cut, bipartite matching - Graph coloring, 4-coloring theorem - Independent sets, vertex covers, edge covers 11. Trees (Graphs) - Necessary and sufficient conditions for a graph to be a tree - Rooted trees, forest, binary trees - Decision trees, game trees, backward induction - Isomorphism 12. Counting (Combinatorial analysis) - Basic techniques (bijections, cartesian products, partitions, binary relations, inclusion-exclusion principle) - Recurrence relations, generating functions - Multi-dimensional Catalan numbers, Chung-Feller theorem	
핵심역량	problem-solving skills, logical thinking, creativity	

제5주 9.15.~9.19.	13. Discrete probability (Combinatorial analysis) • Frequentist/Bayesian probability • Finite probability space • Bayesian decision problems • Probabilistic methods (counting sieve, linearity of expectation, derandomization) 14. Cayley's formula for the number of spanning trees (Combinatorial analysis) • Proof by bijection • Proof by recurrence relations • Proof by double counting • Proof by linear algebra 15. Chains/antichains #1 (Combinatorial analysis) • Hasse diagrams for posets • Decomposition of posets into chains/antichains • Dilworth's theorem (a specialization to Hall's condition for perfect matching)	
핵심역량	problem-solving skills, logical thinking, creativity	
제6주 9.22.~9.26.	16. Chains/antichains #2 (Combinatorial analysis) • Symmetric chains • Sperner system, LYM inequality • Bollobás's theorem, strong systems of distinct representatives 17. Lattices (Algebraic structures) • Lower/upper bounds in posets, meet/join operations • Distributive lattices, complete lattices • Complemented lattices (boolean algebra) • Applications in stable matching 18. Turing machines (Algorithms) • Turing machines • Formal notions of algorithmic problems and algorithms • Church-Turing thesis	
핵심역량	problem-solving skills, logical thinking, creativity	
제7주 9.29.~10.3.	19. Computability (Algorithms) • Universality, the universal Turing machine • Undecidability of the halting problem • Undecidable problems in CS & math • Proving the undecidability by reduction • Rice's theorem 20. Incompleteness Theorems (Algorithms) • Logical systems • FOL, ZFC axioms, PA axioms • Soundness/Completeness/Consistency • Gödel's incompleteness theorems	개천절(10.3.)
핵심역량	problem-solving skills, logical thinking, creativity	
제8주 10.6.~10.10.	추석연휴, 한글날, 재량 휴업일	

주(기간)	주제 및 내용	(비고)
제9주 10.13.~10.17.	중간고사(10.13.~10.16.)	10.17.(금) 정상수업
제10주 10.20.~10.24.	22. Shortest paths: Dijkstra's greedy algorithm (Graphs) • Properties of shortest paths • Proof patterns with loop invariants (proofs with weaker/stronger induction hypothesis by contradiction) • Applications (communication networks, arbitrage) • Negative-weight edges/cycles, longest path problems 23. Shortest paths: Floyd-Warshall's algorithm (Graphs) • Dynamic programming vs. recursion • Inductive formulation of shortest paths • Behavior with negative-weight edges/cycles • Minimum mean-cycle problem	
핵심역량	problem-solving skills, logical thinking, creativity	
제11주 10.27.~10.31.	24. Algebraic path problems (Graphs) • Idempotent semirings (dioids) • Inductive definition of set of paths • Semiring-based dynamic-programming formulation for algebraic path problems 25. Minimum spanning trees (Graphs) • Cuts, crossing edges, light edges, cut/cycle properties • The generic MST algorithm, specialization to Kruskal/Prim's algorithms • Applications (communication networks, TSP, Steiner trees) 26. Matroids (Combinatorial optimization) • Hereditary sets, k-exchange property • Graphic matroids • The generic greedy algorithm on weighted matroids • k-approximation	
핵심역량	problem-solving skills, logical thinking, creativity	
제12주 11.3.~11.7.	27. Flow network (Graphs) • Max-flow min-cut theorem • Ford-Fulkerson method, capacity scaling • Bipartite matching, Hall's condition for perfect bipartite matching • Applications 28. Polynomial-time reductions (Intractability) • Decision problems • Polynomial-time reductions • Standard reduction techniques (simple equivalence, restriction, local replacement, component design) • Reduction examples (3-SAT to INDEPENDENT SETS, 3-SAT to HAMILTONIAN CYCLES, 3-SAT to 3-DIMENSIONAL MATCHING, 3-SAT to GRAPH COLORING) 29. NP-completeness (Intractability) • Certifier-based definition of the class NP • Cook-Levin theorem • How to cope with intractability • Important complexity classes beyond NP (co-NP, PSPACE, ZPP, BPP, RP)	
핵심역량	problem-solving skills, logical thinking, creativity	

제13주 11.10.~11.14.	30. Combinatorial games #1 (Game theory) - Inductive definition of winning/losing positions - Impartial combinatorial games, Nim game - Sprague-Grundy theorem - Partizan games - Surreal numbers 31. Non-cooperative strategic form games (Game theory) - Dominant strategy equilibrium - Pure strategy Nash equilibrium - Mixed strategy Nash equilibrium 32. Mechanism design (Game theory) - Implementation of solution concepts (e.g. dominante strategies, Bayesian-Nash equilibrium) - The revelation principle, truthful implementation - Vickrey-Groves-Clarke (VGC) mechanism - Algorithmic issues in mechanism design	
핵심역량	problem-solving skills, logical thinking, creativity	
제14주 11.17.~11.21.	33. Latin squares (Combinatorial design) - Incomplete Latin squares - Orthogonal Latin squares - Construction by finite fields and extension from prime orders - Applications (design of experiments) 34. Block designs (Combinatorial design) - Regularity, existence - Symmetric block design by using finite fields - Projective planes 35. Modular arithmetic (Cryptography) - Divisibility - Additive/multiplicative group modulo n - Modular linear equations, Chinese remainder theorem - Modular exponentiation, discrete logarithm	
핵심역량	problem-solving skills, logical thinking, creativity	
제15주 11.24.~11.28.	36. Cryptology (informal overview) (Cryptography) - Trapdoor one-way functions - Private-key cryptosystems, sharing private keys - Public-key cryptosystems 37. The RSA cryptosystem (Cryptography) - The RSA cryptosystem, security issues, digital signature - Primality testing (the prime density theorem, the Miller-Rabin randomized testing, the AKS algorithm) - RSA attacks, heuristics for prime factorization 38. Other cryptosystems & protocols (Cryptography) - Merkle-Hellman cryptosystem (intractability of the subset-sum problem) - ElGamal cryptosystem (discrete logs, elliptic curves) - Secret sharing, multi-party secure computations, zero-knowledge proofs	
핵심역량	problem-solving skills, logical thinking, creativity	
제16주 12.1.~12.5.	기말고사 대비 복습	보강 주간
제17주 12.8.~12.12.	기말고사(12.08.~12.12.)	

※ 학사 일정에 따라 강의 계획이 변경될 수 있음